

# Statistical methods for the cross-cultural recognition memory pipeline

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This note specifies the statistical procedures implemented in **Analysis-Scripts-v2/**. Each section names the function(s) responsible.

## 1 Data structure

Each participant  $i$  in group  $g$  contributes one recognition memory session with mixed first-presentation and repeat trials. For each unique stimulus  $j$  encountered we record two binary indicators:

- the *hit* indicator  $h_{gij} \in \{0, 1\}$ , defined on the nonzero-ISI repeat occurrence of stimulus  $j$ , equal to `isResponseCorrect`;
- the *false-alarm* indicator  $f_{gij} \in \{0, 1\}$ , defined on the first (non-repeat) occurrence of stimulus  $j$ , equal to  $1 - \text{isResponseCorrect}$ .

A participant who did not encounter stimulus  $j$  contributes NaN at  $h_{gij}$  and  $f_{gij}$ . The matrices  $H_g, F_g \in (\{0, 1\} \cup \{\text{NaN}\})^{n_g \times m}$  are built by `calculateSplitHalfReliability.m`, where  $n_g$  is the number of participants in  $g$  passing the  $d'$ -at-ISI=0 inclusion threshold and  $m$  is the union of stimuli encountered by any included participant. Subsequent analyses align to the intersection of stimulus sets across the groups being compared.

## 2 Within-group split-half reliability

Reliability of an item-level rate vector is estimated by random split-half over participants (default; `splitDim = 1`) or over stimuli (`splitDim = 2`). For each of  $S$  random splits ( $S = 10,000$  by default), participants are partitioned at random into two halves. Item-level means  $\bar{h}^{(1)}, \bar{h}^{(2)} \in \mathbb{R}^m$  are computed within each half and Spearman's rank correlation  $r_s$  is computed across items, ignoring items where either half is NaN. Items with fewer than three valid pairs are dropped from that split. The half-set reliability is the *median* of the per-split correlations,

$$\hat{\rho}_{1/2} = \text{median}\{r_s^{(1)}, \dots, r_s^{(S)}\}. \quad (1)$$

The median is used in place of an arithmetic or Fisher- $z$  mean because it is robust to skew and to occasional degenerate splits (e.g., a split where one half's item rates have near-zero variance), and it is invariant to monotone transforms of  $r$ . The standard deviation of  $\{r_s\}$  is reported as a dispersion statistic.

The reliability of the full set follows from the Spearman–Brown correction,

$$\hat{\rho}_{\text{SB}} = \frac{2\hat{\rho}_{1/2}}{1 + \hat{\rho}_{1/2}}, \quad (2)$$

and is the quantity used for attenuation correction (§5). Implementation: `estimateSplitHalfFlexible.m`. The FA matrix follows the same procedure.

### 3 Intergroup itemwise correlation: point estimate

Given two groups  $A$  and  $B$  aligned on a shared item set  $\mathcal{S}_{AB}$ , define group-level item means

$$\bar{h}_{Aj} = \frac{1}{|\{i : h_{Aij} \neq \text{NaN}\}|} \sum_{i: h_{Aij} \neq \text{NaN}} h_{Aij}, \quad (3)$$

and analogously  $\bar{h}_{Bj}$ . Item  $j$  is retained if it has at least `minResp` non-NaN observations in each group (default `minResp` = 2); items below this threshold are removed before correlation. At least `minItems` = 5 retained items are required.

The point estimate is

$$\hat{r}_{AB} = \text{corr}_{\text{Spearman}}(\{\bar{h}_{Aj}\}_{j \in \mathcal{V}_{AB}}, \{\bar{h}_{Bj}\}_{j \in \mathcal{V}_{AB}}), \quad (4)$$

where  $\mathcal{V}_{AB} \subseteq \mathcal{S}_{AB}$  are the items meeting the coverage criterion. Spearman is the default given the bounded, often skewed nature of group-mean hit rates.

### 4 Bootstrap confidence intervals

A cluster bootstrap is used to obtain confidence intervals on  $\hat{r}_{AB}$ .

**Participant bootstrap** (`BootstrapDim` = 1, **default**). On each of  $B$  iterations, participants in group  $A$  are resampled with replacement to size  $n_A$ , independently from group  $B$ . Item-level means  $\bar{h}_A^{(b)}, \bar{h}_B^{(b)}$  are recomputed on the resampled participants, the coverage filter is re-applied, and a per-iteration correlation  $r_{AB}^{(b)}$  is recorded.

**Stimulus bootstrap** (`BootstrapDim` = 2). A single set of stimuli is resampled with replacement from  $\mathcal{S}_{AB}$  and used for both groups. Repeatedly drawn stimuli appear as repeated entries in the correlation vector — the standard cluster-bootstrap-by-items quirk.

The headline central value is the sample point estimate  $\hat{r}_{AB}$  (Eq. 4) computed once on the original (un-resampled) data; the bootstrap is used only for uncertainty quantification, not for re-estimating the central value. Across iterations the distribution is summarized by:

- a 95% percentile CI =  $[P_{2.5}, P_{97.5}]$  of the  $r^{(b)}$  distribution;
- the *median* of the bootstrap distribution as a secondary sanity-check value. The bootstrap median should sit close to  $\hat{r}_{AB}$ ; a large gap signals skew in the resampling distribution and is itself diagnostic.

No Fisher- $z$  averaging is used. Fisher- $z$  stabilizes variance of correlations but introduces a transform on the central value; reporting the untransformed  $\hat{r}$  alongside a percentile CI avoids this.

## 5 Attenuation correction

Reliability-corrected (“disattenuated”) correlations follow the classical Spearman correction:

$$\hat{r}_{AB}^* = \frac{\hat{r}_{AB}}{\sqrt{\hat{\rho}_{SB}^A \hat{\rho}_{SB}^B}}. \quad (5)$$

The denominator is floored at machine epsilon and the corrected estimate is clamped to  $[-1, 1]$ . Three modes select which reliability is used inside the bootstrap:

‘fixed’ full-sample  $\hat{\rho}_{SB}^{A,B}$  are used unchanged in every iteration. Fast; conservative.

‘subset’ reliabilities are recomputed once on the items surviving the coverage filter at the point estimate, then held fixed.

‘per-draw’ reliabilities are recomputed on each bootstrap resample, using the same resampled participants/stimuli and the same item subset as that iteration’s correlation. The rigorous choice; propagates resample-level uncertainty into the corrected estimate.

## 6 Paired-bootstrap comparison of three intergroup correlations

The substantive question—whether the US–San Borja correlation differs from US–Tsimane’, etc.—requires testing equality of correlations that *share a group*. Independent bootstrap CIs for each correlation are insufficient:  $\hat{r}_{AB}$  and  $\hat{r}_{AC}$  are dependent (both depend on  $A$ ), and ignoring this dependence overstates the variance of their difference.

We use `pairedBootstrapCompareCorrelations.m`, which preserves dependence by *sharing the bootstrap resample of any group that appears in both correlations being compared*. On iteration  $b$ :

1. Resample participants within each group independently: indices  $\pi_A^{(b)}, \pi_B^{(b)}, \pi_C^{(b)}$  (participant bootstrap), or draw a single shared item resample  $\pi_{\text{stim}}^{(b)}$  (stimulus bootstrap).
2. Compute group means from the resampled units, and obtain all three correlations  $r_{AB}^{(b)}, r_{AC}^{(b)}, r_{BC}^{(b)}$  on the items meeting `minResp` in all three groups. Critically,  $\bar{h}_A^{(b)}$  entering  $r_{AB}^{(b)}$  and  $\bar{h}_A^{(b)}$  entering  $r_{AC}^{(b)}$  are computed from the *same*  $\pi_A^{(b)}$ , and analogously for  $B$  and  $C$ .
3. Form pairwise differences  $d_{AB,AC}^{(b)} = r_{AB}^{(b)} - r_{AC}^{(b)}$ , etc.

The empirical distributions  $\{d^{(b)}\}_{b=1}^B$  are the bootstrap distributions of the differences; their P<sub>2.5</sub>, P<sub>97.5</sub> percentiles give 95% CIs.

We emphasize that “paired” here refers to the *shared-group resample*, not matched participants across cultures; participants are not matched across sites.

## 7 Two-sided p-values

Two p-value definitions are computed and reported.

**Recentered-null (default).** Let  $\bar{d}$  denote the bootstrap mean of  $d^{(b)}$  and  $d^{\text{obs}}$  the observed difference computed from the original sample. The null is constructed by recentering the bootstrap distribution at zero,  $\tilde{d}^{(b)} = d^{(b)} - \bar{d}$ , and

$$p_{\text{null}} = \widehat{\Pr}(|\tilde{d}| \geq |d^{\text{obs}}|), \quad (6)$$

floored at  $1/(B+1)$ . This is the bootstrap analogue of a permutation test under  $H_0 : \delta = 0$  and is the test we recommend reporting.

**Straddle-zero (legacy).**

$$p_{\text{strad}} = 2 \min(\widehat{\Pr}(d^{(b)} > 0), \widehat{\Pr}(d^{(b)} < 0)). \quad (7)$$

Asks whether the bootstrap CI of the difference crosses zero. Biased under skew of the bootstrap distribution, particularly at small  $n$ . Retained for backward comparability with prior pipeline runs.

When (6) and (7) diverge appreciably, the recentered-null variant is the appropriate one to report; the divergence itself signals skew in the bootstrap of  $d$ .

## 8 Small-sample considerations and validation

With per-group  $n \in [15, 35]$  and shared-item counts  $m \in [40, 80]$ —typical for cross-cultural fieldwork—both the bootstrap CIs and the paired test operate at the lower edge of their valid range. We address this two ways.

**Reporting.** Each pipeline run records (i) the per-iteration retained item count after the `minResp` filter, (ii) within-group  $\hat{\rho}_{\text{SB}}$ , and (iii) the gap between (6) and (7). Any of these falling far outside their typical range signals that the test is operating near its limits and the corresponding p-value should be interpreted with caution.

**Calibration via simulation.** The script `powerSimulationPairedBootstrap.m` generates synthetic three-group recognition memory data with a specified true correlation structure across groups and per-group sample sizes matching the real study. Under  $H_0$  (all pairwise true correlations equal), it reports the empirical type-I error rate of each pairwise test; under  $H_1$  (one correlation differs by  $\Delta$ ), it reports power. CI coverage for each difference is also reported. Running this once at the actual site sizes provides the operating characteristics that should accompany any reported p-value from the real data.

## 9 Software and reproducibility

All MATLAB analyses use MATLAB 2018b+. Random splits and bootstrap resamples are drawn with `rng('shuffle')` at the top of each routine; for reproducible runs, set `rng(seed)` once before calling `run_cross_cultural_analysis.m`.

A Python twin lives in `python/` and mirrors MATLAB's `stats/1:1: split_half.py, intergroup_corr.py, paired_bootstrap.py, power_simulation.py`, with shared helpers in `_utils.py`. The top-level driver is `run_cross_cultural_analysis.py`. The two languages compute the same statistics (including the recentered-null and straddle-zero p-values of §7 and the calibration sim of §8) and produce numerically matching results to within bootstrap sampling error. Python 3.7+, NumPy, SciPy.

## A Function-to-equation map

| Function                                        | Equation / Step                     |
|-------------------------------------------------|-------------------------------------|
| <code>calculateSplitHalfReliability</code>      | §1; calls §2                        |
| <code>estimateSplitHalfFlexible</code>          | $\hat{\rho}_{1/2}$ in §2            |
| Spearman–Brown (inline)                         | Eq. (2)                             |
| <code>bootstrapIntergroupCorrelationSEM</code>  | Eqs. (4), (5); §4, §5               |
| <code>pairedBootstrapCompareCorrelations</code> | §6; Eqs. (6), (7)                   |
| <code>plotTripleIntergroupBar_v2</code>         | reporting                           |
| <code>sensitivityMinResp</code>                 | sensitivity to <code>minResp</code> |
| <code>powerSimulationPairedBootstrap</code>     | calibration                         |